

Операционные системы

Основы интерфейса взаимодействия пользователя с системой Unix на уровне командной строки

Радов Максим Иванович

2026-08-31

Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Приобретение практических навыков взаимодействия пользователя с системой посредством командной строки.

- 1 Определить имя и путь домашнего каталога
- 2 Изучить команду ls.
- 3 Выполнить действия с каталогами.
- 4 Получить дополнительные сведения при помощи справки по командам.
- 5 Изучить команду history.

2. Процесс выполнения лабораторной работы

```
miradov@miradov:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intr  
o$ cd  
miradov@miradov:~$ pwd  
/home/miradov  
miradov@miradov:~$
```

Рисунок 1: Путь к домашнему каталогу

```
miradov@miradov:~$ cd /tmp
miradov@miradov:/tmp$ ls
03facf7f-e95b-4648-88aa-49d8f553818a.zip
happd.sock
hsperfdata_miradov
node-compile-cache
par-6d697261646f76
snap-private-tmp
systemd-private-3b438185ccd84d0987488dc2f68cb78a-abrtd.service-cJVcxa
systemd-private-3b438185ccd84d0987488dc2f68cb78a-chronyd.service-QUeJWb
systemd-private-3b438185ccd84d0987488dc2f68cb78a-colord.service-SU0jqg
systemd-private-3b438185ccd84d0987488dc2f68cb78a-dbus-broker.service-QzYssy
systemd-private-3b438185ccd84d0987488dc2f68cb78a-irqbalance.service-jJcTMy
systemd-private-3b438185ccd84d0987488dc2f68cb78a-low-memory-monitor.service-dXWDzw
systemd-private-3b438185ccd84d0987488dc2f68cb78a-ModemManager.service-Dr26vT
systemd-private-3b438185ccd84d0987488dc2f68cb78a-passim.service-l5wz6A
systemd-private-3b438185ccd84d0987488dc2f68cb78a-pcsd.service-zeCgL7
systemd-private-3b438185ccd84d0987488dc2f68cb78a-polkit.service-3Fxiux
systemd-private-3b438185ccd84d0987488dc2f68cb78a-rtkit-daemon.service-qWTQ6j
systemd-private-3b438185ccd84d0987488dc2f68cb78a-switcheroo-control.service-PHRdn6
systemd-private-3b438185ccd84d0987488dc2f68cb78a-systemd-logind.service-fV0oaZ
systemd-private-3b438185ccd84d0987488dc2f68cb78a-systemd-resolved.service-5EV9JL
systemd-private-3b438185ccd84d0987488dc2f68cb78a-upower.service-IBv303
VMwareDnD
vmware-root_893-3988097506
miradov@miradov:/tmp$
```

Рисунок 2: Команда ls


```
miradov@miradov:/tmp$ ls -la
.
..
03facf7f-e95b-4648-88aa-49d8f553818a.zip
.font-unix
happd.sock
hsperfdata_miradov
.ICE-unix
node-compile-cache
par-6d697261646f76
snap-private-tmp
systemd-private-3b438185ccd84d0987488dc2f68cb78a-abrtd.service-cJVcxa
systemd-private-3b438185ccd84d0987488dc2f68cb78a-chronyd.service-QUeJWb
systemd-private-3b438185ccd84d0987488dc2f68cb78a-colord.service-SU0jqg
systemd-private-3b438185ccd84d0987488dc2f68cb78a-dbus-broker.service-QzYssy
systemd-private-3b438185ccd84d0987488dc2f68cb78a-irqbalance.service-jJcTMy
systemd-private-3b438185ccd84d0987488dc2f68cb78a-low-memory-monitor.service-dXWDzw
systemd-private-3b438185ccd84d0987488dc2f68cb78a-ModemManager.service-Dr26vT
systemd-private-3b438185ccd84d0987488dc2f68cb78a-passim.service-I5wz6A
systemd-private-3b438185ccd84d0987488dc2f68cb78a-pcsd.service-zeCgL7
systemd-private-3b438185ccd84d0987488dc2f68cb78a-polkit.service-3Fxiux
systemd-private-3b438185ccd84d0987488dc2f68cb78a-rtkit-daemon.service-qWTQGj
systemd-private-3b438185ccd84d0987488dc2f68cb78a-switcheroo-control.service-PHRdn6
systemd-private-3b438185ccd84d0987488dc2f68cb78a-systemd-logind.service-fV0oaZ
systemd-private-3b438185ccd84d0987488dc2f68cb78a-systemd-resolved.service-5EV9JL
systemd-private-3b438185ccd84d0987488dc2f68cb78a-upower.service-IBv303
VMwareDnD
vmware-root_893-3988097506
.X0-lock
.X11-unix
.X1-lock
.XIM-unix
miradov@miradov:/tmp$
```

Рисунок 3: Команда ls -la

```
miradov@miradov:/tmp$ ls -l
total 3860
-rw-r--r--. 1 miradov miradov 3949730 Aug 31 09:54 03facf7f-e95b-4648-88aa-49d8f55
3818a.zip
srw-rw-rw-. 1 root    root          0 Aug 31 09:52 happd.sock
drwxr-xr-x. 2 miradov miradov      40 Aug 31 10:14 hspcrfdata_miradov
drwxr-xr-x. 3 miradov miradov      60 Aug 31 10:25 node-compile-cache
drwx----- 3 miradov miradov      60 Aug 31 10:08 par-6d697261646f76
drwx----- 2 root    root          40 Aug 31 09:52 snap-private-tmp
drwx----- 3 root    root          60 Aug 31 09:52 systemd-private-3b438185ccd84d0
987488dc2f68cb78a-abrt.service-cJVcxa
drwx----- 3 root    root          60 Aug 31 09:52 systemd-private-3b438185ccd84d0
987488dc2f68cb78a-chrond.service-QUeJWb
drwx----- 3 root    root          60 Aug 31 09:52 systemd-private-3b438185ccd84d0
987488dc2f68cb78a-colord.service-SU0jqg
drwx----- 3 root    root          60 Aug 31 09:52 systemd-private-3b438185ccd84d0
987488dc2f68cb78a-dbus-broker.service-QzYssy
drwx----- 3 root    root          60 Aug 31 09:52 systemd-private-3b438185ccd84d0
987488dc2f68cb78a-irqbalance.service-jJcTMy
drwx----- 3 root    root          60 Aug 31 09:52 systemd-private-3b438185ccd84d0
987488dc2f68cb78a-low-memory-monitor.service-dXWDzw
drwx----- 3 root    root          60 Aug 31 09:52 systemd-private-3b438185ccd84d0
987488dc2f68cb78a-ModemManager.service-Dr26vT
drwx----- 3 root    root          60 Aug 31 09:53 systemd-private-3b438185ccd84d0
987488dc2f68cb78a-passim.service-I5wz6A
drwx----- 3 root    root          60 Aug 31 09:52 systemd-private-3b438185ccd84d0
987488dc2f68cb78a-pcsd.service-zeCgL7
```

Рисунок 4: Команда ls -l

Опции команды ls

```
miradov@miradov:/tmp$ ls -f
.
..
node-compile-cache
hsperfdata_miradov
par-6d697261646f76
VMwareDnD
03facf7f-e95b-4648-88aa-49d8f553818a.zip
systemd-private-3b438185ccd84d0987488dc2f68cb78a-passim.service-I5wz6A
.X1-lock
.X0-lock
systemd-private-3b438185ccd84d0987488dc2f68cb78a-pcscd.service-zeCgL7
systemd-private-3b438185ccd84d0987488dc2f68cb78a-colord.service-SU0jqg
happd.sock
systemd-private-3b438185ccd84d0987488dc2f68cb78a-ModemManager.service-Dr26vT
vmware-root_893-3988097506
systemd-private-3b438185ccd84d0987488dc2f68cb78a-systemd-logind.service-fV0oaz
systemd-private-3b438185ccd84d0987488dc2f68cb78a-upower.service-IBv303
systemd-private-3b438185ccd84d0987488dc2f68cb78a-switcheroo-control.service-PHRdn6
systemd-private-3b438185ccd84d0987488dc2f68cb78a-rtkit-daemon.service-qWTQ6j
systemd-private-3b438185ccd84d0987488dc2f68cb78a-polkit.service-3Fxiux
systemd-private-3b438185ccd84d0987488dc2f68cb78a-low-memory-monitor.service-dXWDzw
systemd-private-3b438185ccd84d0987488dc2f68cb78a-chrond.service-QUeJWb
systemd-private-3b438185ccd84d0987488dc2f68cb78a-irqbalance.service-jJcTMy
systemd-private-3b438185ccd84d0987488dc2f68cb78a-abrt.service-cJVcxa
systemd-private-3b438185ccd84d0987488dc2f68cb78a-dbus-broker.service-QzYssy
systemd-private-3b438185ccd84d0987488dc2f68cb78a-systemd-resolved.service-5EV9JL
.font-unix
```

Рисунок 5: Команда ls -f

```
miradov@miradov:/tmp$ cd /var/spool
miradov@miradov:/var/spool$ ls -l
total 0
drwxr-x--x. 1 root abrt 24476 Aug 31 09:49 abrt
drwx-----. 1 abrt abrt    0 Dec  4 2025 abrt-upload
drwxr-xr-x. 1 root root   66 Oct 23 2025 anacron
drwx-----. 1 root root   18 Oct 23 2025 at
drwx-----. 1 root root    0 Aug  6 2025 cron
drwx--x---. 1 root lp     6 Dec 12 2025 cups
drwxr-xr-x. 1 root root    0 Jul 30 2025 lpd
drwxrwxr-x. 1 root mail  482 Aug 31 09:52 mail
drwxr-xr-x. 1 root root    0 Jan 27 2026 plymouth
miradov@miradov:/var/spool$
```

Рисунок 6: Каталог /var/spool

```
miradov@miradov:/var/spool$ cd
miradov@miradov:~$ ls
Desktop    Downloads  Music      Public     Videos
Documents  git-extended Pictures    Templates  work
miradov@miradov:~$ ls -al
total 24
drwx-----. 1 miradov miradov 370 Aug 31 10:34 .
drwxr-xr-x. 1 root    root    482 Aug 31 09:52 ..
-rw-----. 1 miradov miradov 1078 Aug 31 10:36 .bash_history
-rw-r--r--. 1 miradov miradov 18 Jul 23 2025 .bash_logout
-rw-r--r--. 1 miradov miradov 144 Jul 23 2025 .bash_profile
-rw-r--r--. 1 miradov miradov 677 Aug 31 10:25 .bashrc
drwx-----. 1 miradov miradov 484 Aug 31 10:27 .cache
drwx-----. 1 miradov miradov 360 Aug 31 10:27 .config
drwxr-xr-x. 1 miradov miradov 0 Aug 31 09:53 Desktop
drwxr-xr-x. 1 miradov miradov 0 Aug 31 09:53 Documents
drwxr-xr-x. 1 miradov miradov 0 Aug 31 09:53 Downloads
-rw-r--r--. 1 miradov miradov 334 Nov 14 2025 .emacs
-rw-r--r--. 1 miradov miradov 170 Aug 31 10:34 .gitconfig
drwxr-xr-x. 1 miradov miradov 74 Aug 31 10:31 git-extended
drwx-----. 1 miradov miradov 136 Aug 31 10:02 .gnupg
drwx-----. 1 miradov miradov 20 Aug 31 09:53 .local
drwxr-xr-x. 1 miradov miradov 34 Oct 23 2025 .mozilla
drwxr-xr-x. 1 miradov miradov 0 Aug 31 09:53 Music
drwxr-xr-x. 1 miradov miradov 0 Aug 31 09:53 Pictures
drwxr-xr-x. 1 miradov miradov 0 Aug 31 09:53 Public
```

Рисунок 7: Файлы в домашнем каталоге

```
miradov@miradov:~$  
miradov@miradov:~$ mkdir newdir  
miradov@miradov:~$ mkdir newdir/morefun  
miradov@miradov:~$ mkdir letters memos misk  
miradov@miradov:~$ ls  
Desktop  Downloads  letters  misk  newdir  Public  Videos  
Documents  git-extended  memos  Music  Pictures  Templates  work  
miradov@miradov:~$ rm letters/ memos/ misk/  
rm: cannot remove 'letters/': Is a directory  
rm: cannot remove 'memos/': Is a directory  
rm: cannot remove 'misk/': Is a directory  
miradov@miradov:~$ rm -r letters/ memos/ misk/  
miradov@miradov:~$ rm -r newdir/  
miradov@miradov:~$ ls  
Desktop  Downloads  Music  Public  Videos  
Documents  git-extended  Pictures  Templates  work  
miradov@miradov:~$
```

Рисунок 8: Действия с каталогами

```
'./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/_resources':  
csl tex  
  
'./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/_resources/csl':  
gost-r-7-0-5-2008-numeric.csl  
  
'./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/_resources/tex':  
preamble.tex  
  
'./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/scripts':  
image-report mpv-shot  
miradov@miradov:~$ ls -t  
git-extended Desktop Downloads Pictures Templates  
work Documents Music Public Videos  
miradov@miradov:~$
```

Рисунок 9: Команда ls -R и ls -t

```
miradov@miradov:~$ help cd
cd: cd [-L|[-P [-e]]] [-@] [dir]
    Change the shell working directory.

    Change the current directory to DIR.  The default DIR is the value of the
    HOME shell variable.  If DIR is "-", it is converted to $OLDPWD.

    The variable CDPATH defines the search path for the directory containing
    DIR.  Alternative directory names in CDPATH are separated by a colon (:).
    A null directory name is the same as the current directory.  If DIR begins
    with a slash (/), then CDPATH is not used.

    If the directory is not found, and the shell option `cdable_vars' is set,
    the word is assumed to be a variable name.  If that variable has a value,
    its value is used for DIR.

Options:
  -L      force symbolic links to be followed: resolve symbolic
          links in DIR after processing instances of `..'
  -P      use the physical directory structure without following
          symbolic links: resolve symbolic links in DIR before
          processing instances of `..'
  -e      if the -P option is supplied, and the current working
          directory cannot be determined successfully, exit with
          a non-zero status
  -@      on systems that support it, present a file with extended
          attributes as a directory containing the file attributes

The default is to follow symbolic links, as if `-L' were specified.
`..' is processed by removing the immediately previous pathname component
back to a slash or the beginning of DIR.

Exit Status:
Returns 0 if the directory is changed, and if $PWD is set successfully when
-P is used; non-zero otherwise.
miradov@miradov:~$
```

Рисунок 10: Справка по команде cd


```
PWD(1)                                User Commands                                PWD(1)

NAME
    pwd - print name of current/working directory

SYNOPSIS
    pwd [OPTION]...

DESCRIPTION
    Print the full filename of the current working directory.

    -L, --logical
        use PWD from environment, even if it contains symlinks

    -P, --physical
        resolve all symlinks

    --help
        display this help and exit

    --version
        output version information and exit

    If no option is specified, -P is assumed.

    Your shell may have its own version of pwd, which usually supersedes the
    version described here. Please refer to your shell's documentation for
    details about the options it supports.

AUTHOR
    Written by Jim Meyering.

REPORTING BUGS
    GNU coreutils online help: <https://www.gnu.org/software/coreutils/>
    Report any translation bugs to <https://translationproject.org/team/>

COPYRIGHT
    Copyright © 2025 Free Software Foundation, Inc. License GPLv3+: GNU GPL
    Manual page pwd(1) line 1 (press h for help or q to quit)
```

Рисунок 11: Справка по команде pwd

```
MKDIR(1)                                User Commands                                MKDIR(1)

NAME
    mkdir - make directories

SYNOPSIS
    mkdir [OPTION]... DIRECTORY...

DESCRIPTION
    Create the DIRECTORY(ies), if they do not already exist.

    Mandatory arguments to long options are mandatory for short options too.

    -m, --mode=MODE
        set file mode (as in chmod), not a=rwx - umask

    -p, --parents
        no error if existing, make parent directories as needed, with their
        file modes unaffected by any -m option

    -v, --verbose
        print a message for each created directory

    -Z      set SELinux security context of each created directory to the de-
            fault type

    --context[=CTX]
        like -Z, or if CTX is specified then set the SELinux or SMACK secu-
        rity context to CTX

    --help display this help and exit

    --version
        output version information and exit

AUTHOR
    Written by David MacKenzie.

Manual page mkdir(1) line 1 (press h for help or q to quit)
```

Рисунок 12: Справка по команде mkdir

```
RMDIR(1)                                User Commands                                RMDIR(1)

NAME
    rmdir - remove empty directories

SYNOPSIS
    rmdir [OPTION]... DIRECTORY...

DESCRIPTION
    Remove the DIRECTORY(ies), if they are empty.

    --ignore-fail-on-non-empty
        ignore each failure to remove a non-empty directory

    -p, --parents
        remove DIRECTORY and its ancestors; e.g., 'rmdir -p a/b' is similar
        to 'rmdir a/b a'

    -v, --verbose
        output a diagnostic for every directory processed

    --help display this help and exit

    --version
        output version information and exit

AUTHOR
    Written by David MacKenzie.

REPORTING BUGS
    GNU coreutils online help: <https://www.gnu.org/software/coreutils/>
    Report any translation bugs to <https://translationproject.org/team/>

COPYRIGHT
    Copyright © 2025 Free Software Foundation, Inc. License GPLv3+: GNU GPL
    version 3 or later <https://gnu.org/licenses/gpl.html>.
    This is free software: you are free to change and redistribute it. There
    is no warranty, to the extent permitted by law.

Manual page rmdir(1) line 1 (press h for help or q to quit)
```

Рисунок 13: Справка по команде rmdir

```
RM(1)                                User Commands                                RM(1)

NAME
    rm - remove files or directories

SYNOPSIS
    rm [OPTION]... [FILE]...

DESCRIPTION
    This manual page documents the GNU version of rm.  rm removes each specified file.  By default, it does not remove directories.

    If the -I or --interactive=once option is given, and there are more than three files or the -r, -R, or --recursive are given, then rm prompts the user for whether to proceed with the entire operation.  If the response is not affirmative, the entire command is aborted.

    Otherwise, if a file is unwritable, standard input is a terminal, and the -f or --force option is not given, or the -i or --interactive=always option is given, rm prompts the user for whether to remove the file.  If the response is not affirmative, the file is skipped.

OPTIONS
    Remove (unlink) the FILE(s).

    -f, --force          ignore nonexistent files and arguments, never prompt

    -i                  prompt before every removal

    -I                  prompt once before removing more than three files, or when removing recursively; less intrusive than -i, while still giving protection against most mistakes

    --interactive[=WHEN]
                        prompt according to WHEN: never, once (-I), or always (-i); without WHEN, prompt always

Manual page rm(1) line 1 (press h for help or q to quit)
```

Рисунок 14: Справка по команде rm

История командной строки

```
71 cd /tmp
72 ls
73 ls -a
74 ls -l
75 ls -f
76 cd /var/spool
77 ls -l
78 cd
79 ls
80 ls -al
81 mkdir newdir
82 mkdir newdir/morefun
83 mkdir letters memos misk
84 ls
85 rm letters/ memos/ misk/
86 rm -r letters/ memos/ misk/
87 rm -r newdir/
88 ls
89 ls -R
90 ls -t
91 help cd
92 man pwd
93 man mkdir
94 man rmdir
95 man rm
96 history
miradov@miradov:~$
```

Рисунок 15: Команда history

3. Выводы по проделанной работе

Мы приобрели практические навыки взаимодействия пользователя с системой посредством командной строки.